

# CurveMole User Manual

## CurveMole User Manual — Preview 0.2.1

### Current manual version: 0.2.1

CurveMole is a general one-dimensional curve fitter with a spectrum-first workflow. It is not a crystallographic refinement program. The same engine serves the desktop GUI, Python API, command line, and YAML workflows.

### Workspace

The main window contains the Series/Curve tree, central plot with collapsible residuals, Model/Parameters panel, and optional dockable panels. Multiple curves can be displayed as Single, Overlay, or Waterfall with arbitrary display-only offsets. The active curve is distinct from selected and visible curves.

The coordinate readout follows the pointer and also reports the nearest data point. Original arrays are float64 and immutable. Screen rendering may be downsampled by PyQtGraph, but fitting and export use all valid, unmasked points.

### Import

Supported formats are TXT, DAT, CSV, and TSV. CurveMole detects common delimiters, headers, comments, decimal points, and decimal commas, then displays the detected interpretation before import. Supported organisations are a single x-y pair, one x with multiple y columns, and multiple x-y pairs through the Python API.

`sigma_y` is interpreted as an absolute one-standard-deviation uncertainty. Inverse variance is converted to residual scaling through its square root. A generic weight column directly scales residuals. `sigma_x` is stored but not used by the version 1 optimiser.

### Data Calculator

The non-modal calculator performs addition, subtraction, multiplication, division, x scaling, and y normalisation. Curve-to-curve operations use linear in-

terpolation by default; nearest and cubic are optional. Extrapolation is off by default and outside values become missing. Every operation is recorded and undoable; **Restore original data** is always available.

## Functions and parameters

Built-in peaks are area normalised. Their analytical area and FWHM are reported. Backgrounds include constant, linear, arbitrary-order polynomial, and cubic spline. Spline x nodes are fixed metadata by default; y nodes are ordinary parameters and can therefore be fitted, fixed, bounded, and linked.

Peak creation is graphical: after selecting the peak function, click its centre and drag horizontally from the centre to either half-height edge. CurveMole derives the initial positive height from the local data residual and uses twice the dragged distance as the initial FWHM. Cubic-spline backgrounds are created by clicking nodes directly on the graph; the exact spline is previewed after every point. **Undo point**, **Finish**, right-click to finish, and **Esc** to cancel are available during placement.

Mask mode supports left-click point masks and left-drag intervals. A right-button drag always applies an interval mask directly, without first enabling Mask mode, using the currently selected Mask/Unmask operation and curve target.

Each parameter may be free, fixed, lower-bounded, upper-bounded, interval-bounded, or linked. Intrinsic bounds are always active: widths are positive and pseudo-Voigt  $\eta$  lies in  $[0, 1]$ . Links use restricted expressions and `${parameter.path}` references. Cycles and infeasible constraints are rejected before fitting.

The Function Builder interprets a restricted expression tree and never executes unrestricted `eval`. Available functions include `abs`, `sqrt`, `exp`, `log`, trigonometric functions, `erf`, `minimum`, `maximum`, `clip`, and `where`.

## Fitting

CurveMole version 1 uses nonlinear least squares only. The local solver is the default. Levenberg–Marquardt is used only for an unbounded ordinary least-squares problem. Differential Evolution is an optional initial search and never invents scientific bounds; every free parameter must have finite user bounds.

Global fits concatenate explicitly weighted residuals from all selected spectra. Equal contribution is optional and never silently enabled. Different spectra may have different models, and expression links can cross spectra. A failed sequential fit pauses at the failing curve for manual correction.

## Statistics and uncertainty

Automatic results include N, free parameter count, degrees of freedom, RSS, RMSE, chi-square, reduced chi-square, descriptive  $R^2$ , standard errors, confidence intervals, covariance, and correlation. AIC/AICc/BIC are displayed only for ordinary least-squares results. Absolute `sigma_y` leaves covariance unscaled; relative or absent weights estimate residual variance. Robust losses use a sandwich covariance and clearly recommend resampling.

Advanced analyses are explicit: profile likelihood, parametric Monte Carlo, residual bootstrap, and block bootstrap. Every stochastic result records requested, completed, and failed replicates, seed, settings, and empirical intervals.

## Project and recovery

`.fitproj` is a documented ZIP container containing a versioned JSON manifest, NumPy `.npy` arrays, and SHA-256 checksums. It never contains pickle. Saves are atomic. Modified projects autosave every ten minutes; inactivity does not rotate a recovery. Only the three newest distinct recovery files are retained.

Projects are not encrypted. Use operating-system permissions or encrypted storage when confidentiality is required.

## Privacy and support

CurveMole operates locally, has no telemetry, and uploads nothing automatically. Diagnostic bundles exclude spectra, projects, and results unless explicitly added.

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